AutoEDA Pro — Statistical & Profiling Report

Dataset: **e71bd788-597f-419d-8fde-ddececcbd3db** • Autonomous Machine Learning Telemetry

1. Dataset Overview

Total Rows	0	**Total Columns**	0
Missing Cells (%)	0.00%	**Target Column**	N/A

2. Executive Summary Narrative

Executive EDA & Dataset Summary Report

Target Directory: `C:\Users\Anish Kumar Verma\PycharmProjects\AutoEDA\sandbox_run\e71bd788-597f-419d-8fde-ddececcbd3db`

Processed Files: `agent_plan_log.json`, `agent_state.json`, `current_df.csv`, `metadata_profile.json`, `metrics.json`

Excluded Files: `generated_analysis.py` (Script excluded from summary)

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1. Dataset Overview

- **Dataset Identifier:** `data.csv`

- **Dimensions:** `891` rows x `14` columns

- **Target Variable:** `Survived`

- **Missing Value Columns:** 3

- `Age`: 177 (19.9%)

- `Cabin`: 687 (77.1%)

- `Embarked`: 2 (0.2%)

1.5 Full Column Statistics

Column	Type	Missing %	Unique %	Mean	Median	Std	Skew	Kurtosis
`PassengerId`	`int64`	0.0%	100.0%	446.0	446.0	257.35	0.0	-1.2
`Survived`	`int64`	0.0%	0.22%	0.38	0.0	0.49	0.48	-1.78
`Pclass`	`int64`	0.0%	0.34%	2.31	3.0	0.84	-0.63	-1.28
`Name`	`str`	0.0%	100.0%	N/A	N/A	N/A	N/A	N/A
`Sex`	`str`	0.0%	0.22%	N/A	N/A	N/A	N/A	N/A
`Age`	`float64`	19.87%	9.88%	29.7	28.0	14.53	0.39	0.18
`SibSp`	`int64`	0.0%	0.79%	0.52	0.0	1.1	3.7	17.88
`Parch`	`int64`	0.0%	0.79%	0.38	0.0	0.81	2.75	9.78

`Ticket`	`str`	0.0%	76.43%	N/A	N/A	N/A	N/A	N/A
`Fare`	`float64`	0.0%	27.83%	32.2	14.45	49.69	4.79	33.4
`Cabin`	`str`	77.1%	16.5%	N/A	N/A	N/A	N/A	N/A
`Embarked`	`str`	0.22%	0.34%	N/A	N/A	N/A	N/A	N/A

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2. Data Imputation & Preprocessing

Rules Applied:

- Standardized missing string placeholders ('?', 'NA', 'N/A', 'null') to NaN.

- Numeric columns with skewness > 1.0 or < -1.0 use median imputation.

- Numeric columns with skewness between -1.0 and 1.0 use mean imputation.

- Categorical/String columns use mode imputation with 'Unknown' fallback.

Column	Missing (Before)	Method	Fill Value
`Age`	177	median	28.0
`Cabin`	687	mode	B96 B98
`Embarked`	2	constant	Unknown

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3. Outlier Analysis (IQR Method)

No numeric outlier statistics reported.

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4. Derived Domain Attributes & Composite Metrics

- **`FamilySize`**: Formula: `SibSp + Parch + 1` | Purpose: Total number of family members on board.

- **`IsAlone`**: Formula: `FamilySize == 1` | Purpose: Indicator for passengers traveling without family.

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5. Statistical Hypothesis Testing & Key Predictors

All predictors below were tested against `Survived` and found statistically significant (p < 0.05), ranked by effect size.

Feature	Test Type	Effect Size	Label	P-Value	Why It Matters
`Sex`	ANOVA	0.5434	Large effect	1.4061e-69	Gender is a primary indicator of survival, reflecting historical emergency protocols that prioritized specific groups.
`Pclass`	Pearson Correlation	0.3385	Moderate correlation	2.5370e-25	Socioeconomic status is strongly linked to survival, likely due to the location and accessibility of different cabin tiers.
`Fare`	Pearson Correlation	0.2573	Weak correlation	6.1202e-15	The amount paid for a ticket relates to survival, as higher spending often granted better access to safety resources.
`IsAlone`	ANOVA	0.2034	Large effect	9.0095e-10	Traveling without family is associated with different survival outcomes compared to those moving in groups.

| `Embarked` | ANOVA | 0.1825 | Large effect | 1.3423e-06 | The location where a passenger boarded shows a connection to survival, possibly reflecting different passenger demographics from each port. |

| `Parch` | Pearson Correlation | 0.0816 | Negligible correlation | 1.4799e-02 | The number of parents or children traveling together is linked to survival rates during the evacuation process. |

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6. Redundancy & Multicollinearity Analysis

Numeric-Numeric High Correlation Pairs (|r| >= 0.85):

Feature 1	Feature 2	Correlation (r)	Interpretation
`SibSp`	`FamilySize`	0.8907	Strong correlation

Cross-Type Redundant Pairs (categorical vs. its own numeric/ordinal encoding, Eta >= 0.85):

Categorical Feature	Numeric Feature	Correlation Ratio (Eta)	Interpretation
`SibSp`	`FamilySize`	0.8929	High cross-type redundancy between 'SibSp' and 'FamilySize' (Eta = 0.8929).
`FamilySize`	`SibSp`	0.9173	High cross-type redundancy between 'FamilySize' and 'SibSp' (Eta = 0.9173).

Recommendation: drop one feature from each redundant pair before modeling to avoid multicollinearity.

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7. Generated Visual Artifacts

No PNG/SVG image assets found in directory.

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8. Categorical Associations (Cramer's V)

Feature 1	Feature 2	Cramer's V
`Survived`	`Pclass`	0.3367
`Survived`	`Sex`	0.5426
`Survived`	`SibSp`	0.1874
`Survived`	`Parch`	0.1569
`Survived`	`Embarked`	0.1731
`Survived`	`FamilySize`	0.2857
`Survived`	`IsAlone`	0.2007
`Pclass`	`Sex`	0.1297
`Pclass`	`SibSp`	0.1478
`Pclass`	`Parch`	0.022
`Pclass`	`Embarked`	0.2637
`Pclass`	`FamilySize`	0.2029
`Pclass`	`IsAlone`	0.1275

`Sex`	`SibSp`	0.2059
`Sex`	`Parch`	0.2471
`Sex`	`Embarked`	0.1255
`Sex`	`FamilySize`	0.3128
`Sex`	`IsAlone`	0.302
`SibSp`	`Parch`	0.2399
`SibSp`	`Embarked`	0.0612
`SibSp`	`FamilySize`	0.7645
`SibSp`	`IsAlone`	0.8367
`Parch`	`Embarked`	0.0
`Parch`	`FamilySize`	0.4901
`Parch`	`IsAlone`	0.6858
`Embarked`	`FamilySize`	0.098
`Embarked`	`IsAlone`	0.1118
`FamilySize`	`IsAlone`	0.9961

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9. Predictive Modeling Strategy Blueprint

- **Target Definition:** Survived

- **Problem Type:** Binary Classification

Recommended Algorithms

- Regularized Logistic Regression (baseline)

- Random Forest Classifier

- Gradient Boosting Classifier (XGBoost / LightGBM)

- Support Vector Classifier (SVM)

Feature Selection Strategy

- Exclude high-cardinality ID or text name columns

- Rank features using cross-validated permutation importance and mutual information

- Remove collinear features exceeding correlation threshold > 0.85

Validation Strategy

- Stratified K-Fold Cross-Validation (5 folds)

- Evaluate Balanced Accuracy, Macro F1, Precision-Recall AUC, and Confusion Matrix

Overfitting Risk Mitigation

- Apply regularization penalties (L1/L2)

- Limit tree depth and enforce minimum samples per leaf

- Perform hyperparameter tuning strictly within cross-validation folds

- **Executive Summary:** Target: 'Survived' (Binary Classification). Model recommendations and validation strategy tailored for 891 rows x 14 columns.

Report generated automatically by `summary_generator.py`

4. Statistical Hypothesis Testing Results

Feature	Test Conducted	Statistic	P-Value	Significant
Sex	ANOVA	0.5434	1.4061e-69	**YES (α=0.05)**
Pclass	Pearson Correlation	0.3385	2.5370e-25	**YES (α=0.05)**
Fare	Pearson Correlation	0.2573	6.1202e-15	**YES (α=0.05)**
IsAlone	ANOVA	0.2034	9.0095e-10	**YES (α=0.05)**
Embarked	ANOVA	0.1825	1.3423e-06	**YES (α=0.05)**
Parch	Pearson Correlation	0.0816	0.0148	**YES (α=0.05)**

5. Predictive Modeling Strategy Blueprint

Target: 'Survived' (Binary Classification). Model recommendations and validation strategy tailored for 891 rows x 14 columns.

Task Type	N/A
Recommended Models	N/A
Validation Strategy	5-Fold Stratified K-Fold